

Designed of an Integrated Attendance Management Tool

Matthew Clarke

Supervisor: Josh Flynn

Submitted in partial fulfilment of the requirements for the degree of Bachelor of Software Engineering (Honours)

School of Information and Physical Sciences

The University of Newcastle, Callaghan, NSW 2308, Australia

03/11/2024

Signed: 

Student: Matthew Clarke

Signed:

Supervisor: Josh Flynn

ABSTRACT

Attendance management systems are essential in academic institutions to track student attendance accurately. The traditional manual method however has downsides such as being slow and prone to human error. Modern attendance systems are faster and do not tend to suffer from human error, and there have been many attendance systems that have started to be switched to recently. The University of Newcastle uses GPS tracking to ensure students are physically present and makes the students record their own attendance rather than have an instructor do all the work. It is much faster but is often unreliable and fails to properly record attendance. Hence, the university wishes to develop a new integrated attendance management system to solve the current system's problems.

The report documents the outcomes of the proposed new attendance management system that uses QR codes to record attendance. It encompasses the entire project's lifecycle, including how the project was designed, developed, implemented, tested, and evaluated. The findings from the report reveal that proposed innovative solution is feasible and accurately records attendance with no errors and little additional complications to implementation. The system was tested for both its usability and performance, and they demonstrated that it was highly usable (94 SUS score) and able to handle less than 100 users per second using the system. The project contributes to the field of attendance systems by using a novel approach to prevent fraud with a low maintenance and inexpensive solution, without needing complex authentication methods like biometrics or GPS tracking.

ACKNOWLEDGEMENTS

I would like to thank my project's supervisor Mr. Josh Flynn who guided the project's progress from start to finish, organised meetings and presentations with the project client and stakeholders, and ensured that the project would be completed on time.

I would also like to thank the entire LTS team for giving me this project to work on throughout the year, helping with testing the final project, and being great colleagues to work with.

TABLE OF CONTENTS

1. Introduction.....	1
1.1 Background.....	1
1.2 Problem Statement.....	1
1.3 Objectives	2
1.4 Scope.....	3
1.4.1 Inclusions	3
1.4.2 Exclusions	3
1.5 Significance	3
1.6 Outline	3
2. Literature Review	5
2.1 Scope.....	5
2.2 Current Solutions	5
2.2.1 QR Code Attendance Systems.....	5
2.2.2 RFID/NFC Attendance Systems.....	6
2.2.3 Biometric Attendance Systems.....	7
2.3 Critical Analysis	8
2.4 Additional Tools	9
2.5 Summary.....	9
3. Methodology	11
3.1 Research Design.....	11
3.2 Data Collection	11
3.3 Data Analysis	12

3.4 Validation and Verification.....	12
3.5 Software Development Methodology	12
3.6 Tools and Technologies	14
4. Design	15
4.1 System Architecture.....	15
4.2 Data Management	17
4.3 User Interface (UI).....	18
4.4 Authentication.....	20
3.8 Ethical Considerations	20
5. Implementation and Evaluation	21
5.1 Development Process.....	21
5.2 Challenges and Solutions.....	23
5.3 Evaluation Methods	23
5.4 Evaluation Criteria.....	24
5.5 Testing	25
5.4.1 Unit Testing	25
5.4.3 Usability and UAT Testing.....	26
5.4.4 Performance Testing	26
6. Results and Discussion.....	27
6.1 Presentation of Data.....	27
6.2 Analysis and Interpretation	28
6.2.1 Usability Testing Results	29
6.2.2 UAT Results	29

6.2.3 Load Testing Results	30
6.3 Contributions	30
6.4 Comparison with Literature	30
6.5 Limitations	31
7. Conclusion	32
7.1 Summary	32
7.2 Implications	32
7.3 Recommendations.....	32
References	33
Appendix A. Usability Test Questionnaire 1	35
Appendix B. Usability Test Questionnaire 2	39

LIST OF FIGURES

Figure 1: System Architecture of Prototype 1	15
Figure 2: System Architecture of Prototype 2	16
Figure 3: MVC Design Pattern	16
Figure 4: EER Diagram representing the data objects.....	17
Figure 5: Instructor’s view of attendance data.....	18
Figure 6: Full attendance history shown on desktop devices (left) and mobile devices (right) ...	19
Figure 7: Instructor's view when scanning cards.	19
Figure 8: OpenID Connect Launch Flow (1EdTech Consortium, Inc., 2019)	20
Figure 9: The application used GitHub Actions to run tests on GitHub.....	26
Figure 10: Attendance Data Displayed on Desktop and Mobile Devices	29

LIST OF TABLES

Table 1: Summary of the benefits for each attendance system, synthesised from the reviewed literature, with significant insights drawn from ‘Attendance Management System for Educational Sector: Critical Review’ (Arif et al., 2018).	8
Table 2: Usability Testing Results (September 23)	27
Table 3: UAT Results (October 28).....	28
Table 4: Load Testing Results	28

1. INTRODUCTION

1.1 Background

In many academic institutions, attendance records are kept to assess students and analyse student engagement in courses. There is often a minimum attendance requirement that students need to meet to pass a course which means that there needs to be a method for recording attendance and a system for keeping the attendance records over time. Attendance management systems are software developed for managing daily student attendance in academic institutions (Shoewu & Idowu, 2012). They allow students to access to their attendance records for any class and for instructors or administrators to evaluate a student's attendance history and generate a report for an entire class.

The main difference between attendance management systems is the method used for taking attendance. The main conventional method used for taking attendance in schools is where the teacher calls the names of students from a role call list and marks them as either attended or not after students respond to their name being called (Singh et al., 2015). An alternative conventional method is where students cross their name off themselves when they walk into the classroom. The alternative method is often quicker than the main conventional method but both methods require the hand-written attendance data to be transferred later onto a computer for safer storage and backup.

The conventional methods are often tedious and slow, which is why many academic institutions have tried to switch to a more modern approach that utilises newer technologies. These modern attendance systems aim to streamline attendance processes and provide both students and instructors with immediate access to their attendance records. The most used attendance methods in modern attendance management systems include scanning barcodes or QR codes on an ID card or a mobile phone, scanning RFID or NFC tags through a specialised reader, or identifying a student through the analysis of their biometric data such as facial or fingerprint recognition (Arif et al., 2018). These methods can also be combined with other technologies, such as tracking a user's location while scanning a barcode. The combination of methods adds a second layer of security for authenticating the correct user but also makes the process more complicated and longer, so a balance between speed and security is required.

1.2 Problem Statement

The University of Newcastle already has a modern attendance management system in place that has benefits over conventional attendance methods. They currently have an attendance tool in the 'myUni' app, which is the app used by the University of Newcastle as a central source for all other university tools and requires students to be logged in before using them. The attendance tool is a third-party software called 'campusM' which tracks student's locations whenever a student wants to record their attendance, and only records attendance successfully if the student is on the university campus. The current system is quick when it works because it requires no input from the instructor and students only need to tap a few buttons, however the university has identified some key issues with it:

- The location tracked is sometimes recorded as invalid or incorrect when a student is on campus, which requires the instructor to record the student's attendance manually for them either on paper or on the instructor's attendance app.
- Students may have location access disabled which again requires instructors to manually record the student's attendance.
- Students can bypass the system by either recording attendance outside the classroom but still being on campus or by getting another student to sign in as them and record their attendance for them.
- Attendance history is limited to the last four weeks, so students are unable to see their full attendance history and need to remember which classes they attended to know if they are meeting the attendance requirements.

Having identified these problems with the current attendance management system, the University of Newcastle wants to improve their attendance tool by using another method to record attendance. They also want the new attendance management system to be moved from the 'myUni' app to Canvas LMS (or just Canvas), which is learning management platform used by the university. They store all course data and grades on Canvas already and would like to have the course attendance data their too for convenience. Students use Canvas frequently to check their grades, and having a new integrated attendance management system there will make it much easier for students to see their full attendance history. The new attendance management system should still be quick at recording attendance but also not have many of the errors mentioned above.

1.3 Objectives

The aim of this project is to develop a new integrated attendance management system that is faster and more reliable than the current attendance management system used by the University of Newcastle. This new attendance management system should fulfill the following objectives:

- Automatically log and update attendance records on Canvas once a student marks their attendance in a physical classroom. This will remove the administrative load required to maintain attendance records and make it easier for students to monitor their attendance.
- Produce reliable and accurate attendance logs, minimising the need for instructors to manually enter a student's attendance and thereby allowing for more time in class and reduce human errors.
- Prevent students from recording attendance without being physically present, resulting in a noticeable increase in class sizes and engagement.
- Provide instructors with an easy-to-use attendance tool to view a summary of their course's attendance and update student attendance records when needed.
- Develop a user-friendly interface for both students and instructors on desktop and mobile devices.

1.4 Scope

1.4.1 Inclusions

This project is focused on the development of a fully integrated attendance management system capable of both recording attendance and display full attendance histories to students and instructors. The system will offer a modern way of recording attendance, which will be via an instructor scanning QR codes on a student's unique digital attendance card stored on their phone. The instructor will also be able to manually record attendance if a student forgot their phone, or overwrite student attendance records in case the student forgot to record attendance altogether. To integrate this with Canvas, the entire attendance management system will be developed as an LTI tool so that users can access the system from within Canvas and not have to navigate to an external application (Murray & Feinberg, 2020). The system will also be designed for maintainability by administrators, which will include automatically updating attendance records each week for students that missed their classes, and logging attendance record updates for monitoring and full record keeping.

1.4.2 Exclusions

The new attendance management system will be solely focused on providing the university with an alternative attendance tool that they can use in the future, but it will not interact with the other systems used by the university. For example, the attendance tool uses test student, course, and timetable data rather than interacting with real systems to get this data such as 'myTimetable', which contains a student's entire timetable for their classes each week. The new system also does not handle user login and authorisation since this is handled by Canvas before the user access the LTI tool, however the new system does still enforce basic user access on each page that a user might try and access. There will also not be an option implemented to allow students to record their attendance for online classes since a new attendance method would be needed, however the new system could still be used to store their attendance history when they do record attendance.

1.5 Significance

The impact that the new attendance management system will have on the university is significant for students, instructors, and administrators. For students, they will be able to easily monitor their attendance grades throughout their courses and have full transparency and accountability. For instructors, they will not need to manually record student attendance as much, which will make recording attendance more efficient and save time for teaching students. For administrators, they will not need to update attendance data as much, because instructors will have more control over changing student attendance records when needed without having to contact administration.

1.6 Outline

This report outlines the entire development of the new attendance management system from start to finish and how the research throughout its development has influenced the project. The structure of this report is as follows:

- Section 1 – Introduction: An overview of the project including a brief background to attendance management systems, the problem statement, and the project's objectives, scope, and significance.
- Section 2 – Literature Review: A summary of the literature and evaluation of solutions that have been tried in the past to solve the problem of recording attendance quicker.
- Section 3 – Methodology: A description of how the research was conducted, how the project was developed, and the tools and technologies used for project management.
- Section 4 – Design: An overview of the system architecture and design decisions made during the project. It also explains how the database was designed and what the user interface looked like.
- Section 5 – Implementation and Evaluation: Documents the development process, implementation details, and how the system was both tested and evaluated to be a success.
- Section 6 – Results and Discussion: A presentation of the results from non-functional testing and an assessment of the application's strengths and weaknesses from the results.
- Section 7 – Conclusion: A summary of the key findings from the project's development, the implications of it for future research, and improvements that can be made next.

2. LITERATURE REVIEW

With the increasingly important need for attendance to be streamlined and be as quick as possible, this section reviews the current landscape of attendance management systems so that their benefits and costs can be compared before choosing the solution for this project. In particular, attendance management systems in the educational sector are extremely relevant to this project, and the benefits that are of most interest are efficiency, reliability, cost, security, and ease of use. This is because the system needs to be able to be used by thousands of students at the University of Newcastle and those interests are key in solving the problems with the current attendance system.

2.1 Scope

The literature review focuses on evaluating alternatives to the current attendance management system used by the University of Newcastle. There was a particular focus on attendance management systems that have been previously implemented in academic institutions, though solutions that have been tried in other fields were also be considered. The modern methods for recording attendance that were analysed include QR code systems, RFID and NFC systems, and biometric systems (Ali et al., 2022).

Literature that highlights benefits to system usability, accuracy, or reliability, and reductions in administrative workload or total time to record attendance, as a result of switching to a new attendance system are highly relevant to the objectives of the project and were sought after in the literature review. The following sections will explore the current solutions and existing attendance methods tried in other literature, an analysis and evaluation of the different attendance methods, a few technologies that can assist when recording attendance, and a summary of the findings and what gap this report will try and address.

2.2 Current Solutions

There are three major strategies for validating attendance in the digital landscape. These are QR code based, RFID or NFC based, and biometric based strategies (Ali et al., 2022). They have been created to reduce the time required to record attendance compared to the conventional manual method of recording attendance.

2.2.1 QR Code Attendance Systems

QR codes are an effective way to record attendance since they are cheap to implement, easily accessible from any device with a camera, and can store vast amounts of information (Moshayedi et al., 2021). They have been commonly used to share links, process feedback, and store unique information. There are several ways that QR codes can be used record attendance.

One of the most obvious ways to use QR codes in a classroom setting is to have the instructor generate a random QR code and have the students scan it on their phone (Liew & Tan, 2021).

Liew and Tan attempted to use this method as a replacement to using a five-digit random code generated by the instructor, because it was easy for students to send it to their friends. They wanted to use QR codes coupled with location tracking to solve this problem. However, their system had a few problems. Firstly, it was only accessible from Android devices, and they found that the GPS detection on some phones wasn't very sensitive, which required students to retry multiple times (Liew & Tan, 2021).

Masalha and Hirzallah used this same method for a smart attendance system (Masalha & Hirzallah, 2014), but their system was more advanced because it required the students to log in to a mobile app and captured a photo of the student's face upon opening the app. The system was successfully implemented, although they did record any results on whether the attendance was faster, other than declaring that the process would take less than a minute for any student (Masalha & Hirzallah, 2014). The system is similar to the University of Newcastle's current system since it requires students to both log in to the app and their location to be tracked, so it will still run into the same problems of invalid or inaccurate locations. There is still the chance that the QR code could be sent to another student outside the classroom, and it requires a bit of input from the instructor, who will either need to generate a QR code once at the start of the class, or have it in the corner of every lecture slide, which requires more input from the instructor but allows students to record attendance at any time of the class.

The second main way of using QR codes to record attendance is to have the QR codes on the student's phones and have them scanned by the instructor. Talip and Zulkifli used this way with success to implement a mobile attendance system for a university (Talip & Zulkifli, 2018). They chose to use QR codes because the cost of implementation is much cheaper compared to other technologies like RFID. Their version required students to generate their own unique QR code each time they wanted to record attendance. Overall, they concluded that the new QR code system was much easier and faster, though they did not gather results on the benefits in terms of usability and user experience of the mobile application (Talip & Zulkifli, 2018).

This method was also used by Wenny, Suhartano, and Parenreng to implement a full attendance management system (Wenny et al., 2022). The lecturer begins recording attendance by clicking a button to open a camera for scanning, and the students see their QR code and a schedule of their classes. Their system had a performance of Grade B (average), reliability of 99%, and usability of 88.05% (very good) according to the standards they compared against such as the Telcordia standard and Likert scale. They concluded that it was a success but required an additional authentication factor like facial recognition to prevent data falsification (Wenny et al., 2022).

2.2.2 RFID/NFC Attendance Systems

RFID and NFC technologies are more modern and innovative than QR codes. They have a similar concept to QR codes because they also require a reader (comparable to the camera) and a unique card (comparable to the QR code). With RFID and NFC technologies, radio frequency is used to communicate and share information between the card and reader. The main difference between the two is that NFC uses one device as both the tag and the reader, whereas RFID requires one device for each functionality (Moshayed et al., 2021). It tends to be quicker than

QR codes because you don't need to aim the phone at the card, you just need them within close proximity. However, they require specialised devices to operate and are a bit more expensive.

Qureshi used RFID technology for a university attendance system to replace their previous method of printing attendance sheets from the electronic system and recording attendance manually (Qureshi, 2020). They concluded that it saved time and reduce the chance of errors, but RFID reader hardware is needed in all classrooms and all students need an RFID tag, so there is a higher initial cost and more initial setup needed. A very similar system was implemented by Lim for attendance systems that could be used by multiple different universities. The tag ID needs to first be stored in memory, and then the students card needs to be within 5cm from the RFID reader. They claimed that their RFID implementation was cheaper than other products on the market and is easily portable, but it still has the problem of needing RFID readers in every classroom, and requiring every student to have a physical card, which is quite small and easy to lose. The RFID reader also has a very small range of detection, especially if the card is not parallel with the reader.

NFC is a specialised subset of RFID technology that is more convenient, has a low energy requirement, and does not require an authentication link between two devices (Dixon & Abuzneid, 2020). Smart phones can act as a NFC device if they have NFC enabled, which is why it is much more convenient. Dixon and Abuzneid used NFC student ID cards to record attendance, where the reader sends the ID number to a cloud-based database. Similarly, Rokade, Shetty, and Wankhade also used NFC technology to create an attendance management system. The student records their attendance by putting their NFC tag against an instructor's mobile phone and the tag is validated before updating the attendance record. The advantage of this approach was that no reader is needed like with RFID, and there was no chance of losing the attendance data, since they used to record attendance manually. This means that there is a much smaller upfront cost, but like with RFID, the university needs to distribute NFC tags to all students before the system can be used.

2.2.3 Biometric Attendance Systems

Biometric attendance systems are much less susceptible to fraud and security vulnerabilities, as they are a measurement of a person's unique physical and behavioural characteristics. While it is common for an attendance system to be based around some other method and then coupled with biometrics for added security and authentication, there are some purely biometric attendance systems that have been tried. Fingerprint recognition was the biometric of choice for the development of an attendance system at a tertiary institution for non-academic staff (Adewole et al., 2014). The proposed system saves time, reduces administrative workload, ensures security of staff records, and eradicates the chance of faking attendance compared to the traditional manual method. This attendance system makes sense for non-academic staff since there are a lot fewer users, but implementing this for an entire university would be much more expensive to set up.

Shoewu and Idowu also used fingerprint recognition for an attendance system to be used by 80 school students in class where fingerprints are captured when enrolling in a course, and compared it with their previous method of manually recording attendance (Shoewu & Idowu, 2012). They found that recording attendance using fingerprint recognition was roughly five times

quicker than manual entry on average. However, the system did occasionally get false rejects before later recognising the student's fingerprint and recording attendance.

Aside from fingerprint recognition attendance systems, iris recognition can also be used although it is less common. Kadry and Smaili implemented iris recognition for a wireless attendance management system and used a sample of 300 people and tested each person's iris four times to check for a match (Kadry & Smaili, 2010). While the system met the needs for daily attendance management, it had a rejection rate of 9.2% and recommended that other methods such as RFID technology or retinal biometrics should be used to improve the reliability of recognition.

2.3 Critical Analysis

Each attendance system has its own strengths and weaknesses. Generally, manual and QR code-based attendance systems are simple and easy to implement, RFID and NFC attendance systems are harder to implement but still easy to use and convenient, and biometric attendance systems are the most expensive but also great at preventing fraud and authenticating students.

To assess which attendance systems are the most suitable for the integrated attendance management tool for the University of Newcastle, there needs to be a set of criteria to compare them against. The criteria that is essential to measure include speed of recording attendance, reliability, the probability of unsuccessful attempts, the cost to implement the system in both time and money, the work needed to get it being used once implemented, and how well the system prevents fraud. Table 1 summarises the benefits and limitations of each system.

Table 1: Summary of the benefits for each attendance system, synthesised from the reviewed literature, with significant insights drawn from 'Attendance Management System for Educational Sector: Critical Review' (Arif et al., 2018).

Attendance system	Quick to record attendance	Reliable results	Inexpensive	Minimal setup needed	Prevents fraud
Manual			✓	✓	
QR	✓	✓	✓	✓	
RFID	✓	✓			✓
NFC	✓	✓		✓	✓
Biometrics	✓				✓

Overall, modern attendance systems are much better than traditional manual attendance systems. The only exception is biometric systems, but that is because a university attendance system has thousands of students and fraud prevention is not the highest priority. QR code, RFID, and NFC based attendance systems are all great options for a university attendance system with advantages and disadvantages of all three.

NFC attendance systems tend to be better than RFID attendance systems if it can be implemented because the instructor's phone can be used as the reader without needing a physical card reader in every classroom. They are very convenient and require minimal setup, but they are

more expensive and harder to implement than QR code attendance systems. The downside to QR code attendance systems is that they do not prevent fraud on their own, however there are options that when combined with QR codes make them more secure, such as rotating the barcode at a set interval, requiring users to log in before scanning, or tracking locations once scanned.

2.4 Additional Tools

The option chosen to solve the problem of QR codes on their own allowing students to bypass the system was the rotating barcodes option. These are tough to implement because you need the server and the QR code to quickly synchronise with each other to know what the valid QR code is at the time of scanning it. For this reason, it is easier to use an external application that has already implemented the feature, which handles QR code generation and synchronisation. One such application that has already implemented this is Google Wallet, which is one of the most common mobile wallets.

Mobile wallets like Google Wallet are versatile applications that store many kinds of digital cards, including generic ones that can be customised by verified businesses. They have been slowly becoming more popular because the cards are convenient to use, less likely to be lost than physical cards, and easily stored in one place (Shin, 2009). They have seen use in attendance management systems because universities can sign up, make their own cards and distribute them to students.

Transact eAccounts is an application used by more than 1100 universities worldwide, including one university in Australia (Monash University's M-PASS). It allows students to add their campus cards into Apple Wallet or Google Wallet and use it to perform any action they normally would with their physical card, including recording attendance. However, third-party services may have issues around security and trust, especially if the information stored in the cards is sensitive (Eng, 2022).

Since the new attendance management system also needs to be integrated with Canvas, the system can be developed as an LTI tool. LTI tools enable seamless integration between the learning system (Canvas) and any external system (the attendance management system) so that the external system can be accessed from a controlled avenue (Murray & Feinberg, 2020). Canvas allows any administrator to connect an external application as an LTI tool for both students and instructors to access. Overall, the easier access will mean that students will use the attendance system more often (Murray & Feinberg, 2020).

2.5 Summary

This literature review has examined various attendance management systems used in the educational sector, including QR code-based attendance systems, RFID/NFC attendance systems, and biometric attendance systems. The advantages and disadvantages of each were analysed, as well as the practical implications they had after implementing them. QR code systems are quick, reliable, and inexpensive, RFID/NFC systems are even faster and reliable but

more expensive and harder to implement, and biometric systems are great at preventing fraud but the most expensive and complicated to implement.

The review identified a gap in the literature, as most QR code-based attendance management systems had addressed students bypassing the system by using additional security factors like GPS tracking or biometrics, but few had attempted to keep the system solely using QR codes while also solving the problem. Rotating barcodes are one of the most secure options to add to the standalone QR code solution, and this area should be further explored to find out if it successfully addresses the problem.

Additionally, additional tools were investigated that will be needed to integrate the application with Canvas and implement rotating barcodes, which were LTI tools and Google Wallet. Both have been used already by other universities around the world to assist in managing academic-related information.

In summary, the literature review highlights the gap in literature that this project intends to fill. The proposed solution is innovative and aims to provide a more efficient, reliable, and secure attendance management system while also being inexpensive and convenient to use.

3. METHODOLOGY

3.1 Research Design

The research method used to investigate the project's effectiveness as an attendance management system is a mix of both quantitative and qualitative measurements. The quantitative aspect of the research involves gathering load test metrics through performance testing, recording the time taken to complete tasks, and user satisfaction scores when conducting UAT and usability testing. The qualitative insights into the project's success include both written and verbal feedback from testing with potential end-users of the system, as well as from general feedback obtained throughout the project's development from the project client and stakeholders.

Quantitative measurements from simulating attendance recording for multiple class sizes would have also been a good insight to get, but since the main purpose of the project was to eliminate the errors with the current attendance management system and integrate it with Canvas, the actual time to record an entire class isn't as important. There was also no baseline for the current attendance management system that was given to us beforehand, so we wouldn't have had anything to compare it with if we did measure the time to record attendance for an entire class.

3.2 Data Collection

Qualitative data was conducted through three main areas. The first is from routine project events such as meetings, presentations, and demonstrations, where the project client, project stakeholders, and project supervisor would give both positive feedback and improvements on the current system and potential paths forward with the project. During project meetings, we had a meeting agenda made so that we knew what the meeting was about, and any questions that were raised and left unanswered were recorded on Confluence for the next meeting. At the end of presentations and demonstrations, a Q/A session was completed for feedback.

The other two ways qualitative data was obtained was by conducting usability testing and UAT. For usability testing, there were two students who completed the testing and spoke aloud while they were using the system. Their thoughts during the testing and at the end when they completed the questionnaire were used as qualitative data, as many of the questions were open-ended to gain more feedback. For UAT testing, feedback was also gathered while users were completing the tests.

However, the questionnaire used for UAT testing was more formal with only questions that required an answer from 1-5. This gave more quantitative results so that an overall understanding of the system's effectiveness could be gained. Finally, Artillery was used to simulate hundreds or thousands of users simultaneously attempting to use the system at the same time. Artillery gives quantitative data every 10 seconds of the test to record the number of users, the response time, the number of failures, and many other statistics.

3.3 Data Analysis

Descriptive analytics produced from load testing were analysed to check if the system is performing at a reasonable and acceptable level. Generally, the system should be able to load attendance data for any user within 5 seconds and not timeout, so that was used as the bar for an acceptable result. Anything greater than that means that the system needs to be optimised more to load attendance data faster for more users.

For usability testing, several questions had benchmarks that would make the system be considered acceptable. We wanted it so that the system was free from errors when recording attendance, so if either of the tests resulted in errors, that would be considered a problem, even if it was just one error. For usability and user satisfaction scores, a score of 4/5 is considered acceptable, and anything below that means that the system has serious problems with the user interface that is affecting user experiences.

For UAT, the average SUS scores were calculated from each questionnaire. Since the average SUS score is 68, anything above that was considered acceptable, however we were aiming for above 80 which makes the application at an A grade level (Lewis, 2018).

3.4 Validation and Verification

Usability testing and UAT was made reliable by making the test environment consistent for all participants. The surveys used in each were the same for each test participant, and when a group of participants were tested at the same time, each participant was isolated from one another so that they didn't have more knowledge of the system than they should before their test. Specifically for UAT, the System Usability Scale was used because it is reliable and an established method for testing that has been used by many others,

Reliability was further maintained when using Artillery for load testing by ensuring that each test was run under the same environment. If you don't run them under the same environment, the test results will be completely different even though the test is the same, because of external factors. The tests were repeated three times each to make sure that the result was accurate. Finally, unit and integration testing help ensure that the system was free from any obvious bugs, and a CI/CD pipeline was set up to have the unit tests run every time code was committed, which helped keep track of old code and make sure that no new bugs had been introduced over time.

3.5 Software Development Methodology

The agile methodology was selected for this project because of its emphasis on continuous improvement through iterative development and consistent feedback with the client and stakeholders. The agile methodology has four core values:

1. **Individuals and interactions over processes and tools.** Processes and tools are important and a proven method for structuring and organising a project, but the agile methodology places more emphasis and value on the people involved during the project's

development, and the effectiveness in which they communicate, collaborate, and work as a team.

2. **Working software over comprehensive documentation.** Documentation can provide a clear outline and big picture of the project; however it requires a large amount of time and resources that diverts attention and effort from actual project implementation. Moreover, these extensive documents are rarely used beyond the initial requirements gathering phase, and are often outdated once the project requirements change. The agile methodology states that project progress is measured through working software since it is less ambiguous, and the client and stakeholders can easily answer whether the current prototype meets their needs.
3. **Customer collaboration over contract negotiation.** Contracts are great for specifying important constraints such as timelines and costs, and ensuring that a minimum viable product is developed according to what was agreed upon. However, customer needs often change, and the idea that a client has in mind may not be the idea that the development team have in mind. The agile methodology ensures that a prototype can be quickly developed so that the client can immediately know whether the project is heading in the right direction, and it allows the project to improve and evolve after receiving feedback to align more with the customer's needs.
4. **Responding to change over following a plan.** Traditionally, projects have involved a detailed plan that will be followed from start to finish. However, there are unforeseen problems that will inevitably arise which could create setbacks. If the requirements change, it will lead to the addition or cancellation of features, or changes in the amount of time necessary for implementing features, ultimately leading to the rigid plan becoming obsolete. The agile methodology encourages flexibility so that the once new information is received or the system is better understood, the development team can adjust their approach without huge delays (Al-Saqqa, Sawalha, & AbdelNabi, 2020).

The agile methodology and its four values work well for us because the project:

1. **Has a small team.** This project was primarily developed by junior software developer with the help and guidance of a project supervisor. The use of a highly structured approach would cause unnecessary project management overhead. Using the agile methodology, we have still used tools for project management such as Jira and maintained a Gantt Chart, but we've stayed flexible and kept the ability for the project plan to be changed depending on the project's progress. Through consistent communication between the project manager and supervisor, and frequent updates with the project client and stakeholders, we've been able to successfully convey the progress of the project and update the project requirements upon feedback and discussion, without dedicating lots of effort to writing extensive documentation.
2. **Requires quick delivery of features.** The project solution that we went for was a solution that we weren't sure whether it was feasible because it used new technologies. This meant that it was essential that the features that required the use of new technology had to be developed as quickly as possible, as if it were not feasible, the project solution would need to be reconsidered and a lot of progress would be lost. By using the agile methodology, we were able to verify whether the solution was appropriate before continuing the rest of development. We also had a deadline to complete a proof-of-

concept prototype, which meant that we needed to begin implementing features early on and keep enough time to finish a working prototype on time.

3. **Has uncertain customer needs.** The description given for this project was vague, as for example the description mentioned that the new project would need to integrate with Canvas LMS and save time recording attendance, but the data sent to Canvas LMS or the method to be used to save time were unspecified. The agile methodology allowed us to present our solution and interpretation of the requirements, and then gain more specific feedback from the client and stakeholders about what specific information they wanted to be displayed on Canvas LMS. We were also able to have meetings with the client and stakeholders about alternative solutions, weigh their pros and cons, and why the solution we proposed would work best for the problem.
4. **Has changing requirements.** The project's scope was left open to changes since the number of features in the prototype would depend on the amount of time spent in development. We focused on iterative development and continual improvement of the software, and made sure that the project could be easily adjusted to unforeseen changes in the requirements.

3.6 Tools and Technologies

Several tools and technologies were used to assist in software design and the agile methodology:

- Jira: used as the project management tool.
- Confluence: used to brainstorm ideas and share design documentation.
- Excel: used to create the Gantt Chart.
- Miro: used to create UI mock-ups and wireframes.
- diagrams.net: used for creating architecture diagrams and EER diagrams.
- Excalidraw: used to create AWS architecture diagrams and database diagrams.

4. DESIGN

Design considerations and architectural choices were fundamental to the project's final prototype being successful. The system architecture, database design, and user interface all underwent significant changes throughout the development so that the application could be maintainable for further improvements and user-friendly. The following subsections give a detailed description of the design considerations as well as ethical issues that were considered throughout the project.

4.1 System Architecture

The system architecture initially was monolithic built using vanilla JavaScript on the frontend, Express.js on the backend for the API endpoints, and PostgreSQL for the database. The application was developed as an LTI tool because it means that user login and course selection are handled by Canvas as the identity provider rather than the application itself. The application receives these details from Canvas once the LTI tool is launched so that the correct view is rendered, and access control can be enforced.

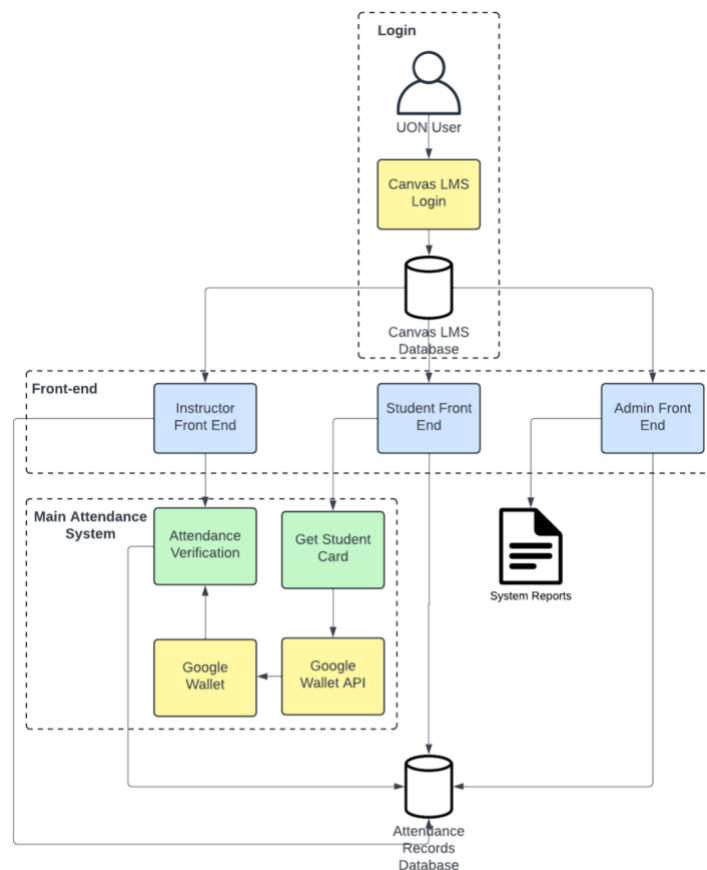


Figure 1: System Architecture of Prototype 1

The first prototype was a proof-of-concept prototype, meaning that it was made to verify whether the proposed solution was a valid solution to the problem. However, the monolithic architecture was unmaintainable, and the backend had several security problems because I was unfamiliar with the Open ID Connect (OIDC) third-party flow used by Canvas LMS for authenticating LTI tools. For this reason, the code was redesigned as a component-based architecture using React for the frontend and the same technologies as before for the backend.

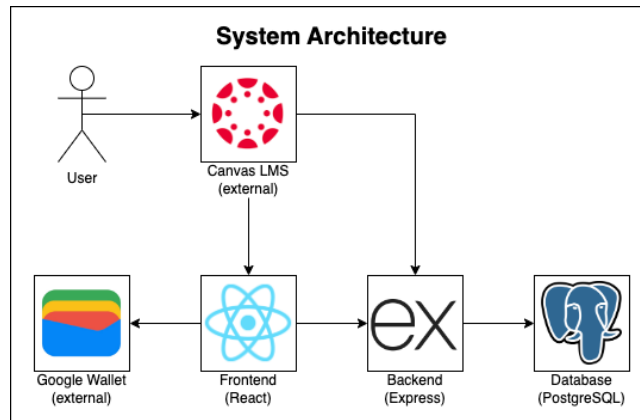


Figure 2: System Architecture of Prototype 2

The backend code was also hard to follow because the files were extremely large. The application was refactored to use a MVC design pattern where:

- The models represented the data and business logic of the application, interacting with the PostgreSQL database.
- The views represented the presentation layer of the application and render the React components.
- The controllers act as an intermediary between the models and views by handling incoming HTTP requests from the frontend as routes, processing them and interacting with the models.

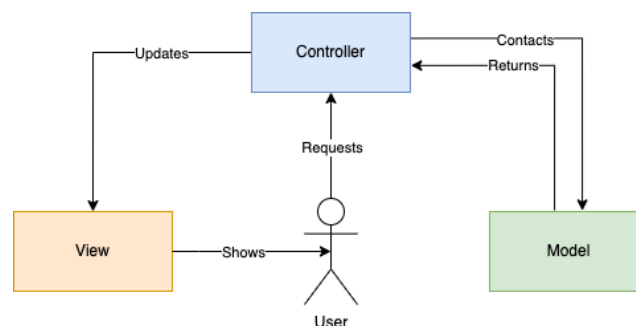


Figure 3: MVC Design Pattern

The MVC design pattern helped with separating the application into individual units of code that had minimal overlapping in the functions they were concerned with. It also made it much easier

to detect errors when unit testing the code, because the code could be tested as one unit rather than needing to use multiple functions in each test.

4.2 Data Management

Attendance records are stored in a PostgreSQL database where each attendance record represents a single attendance period. This means that for a single student, they will have attendance records for every teaching session they are enrolled in, and each record represents a single week. The attendance records are created at the start of each semester or trimester and set to either 'Not required' or 'NULL' and then the status is changed whenever a student records their attendance. This approach makes it easy to look up any attendance record, because each record contains fields such as 'enrolmentId' and 'week', and each enrolment has fields such as 'userId' and 'teachingSessionId'. The advantage of having all attendance records initially created already means that it is easy to update the attendance records. Since no new records need to be created, the only updates need is to the 'status' field.

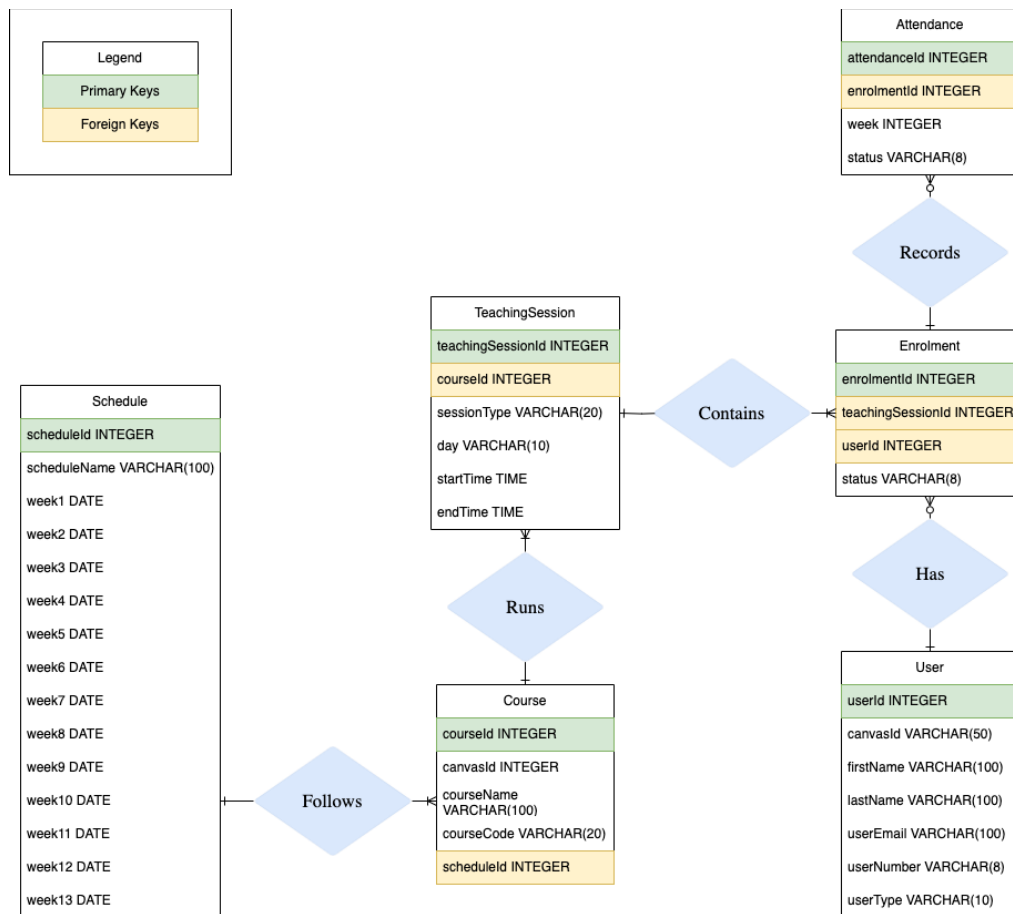


Figure 4: EER Diagram representing the data objects.

The database also keeps track of user data such as names, emails, numbers, and types, and course data such as names, course codes, and schedule types. The system also needs to know the times

and dates to record attendance for, since the system is not integrated with the University of Newcastle's timetabling system. This means it keeps track of schedules, which are either semesters or trimesters, and contain the dates of every week in the schedule, as well as teaching sessions, which are either lectures, labs, tutorials, or workshops, and contain the day they are on and both the start and end times of the session.

The attendance management system does not keep track of the digital attendance cards and what their QR codes are. Normally, you would need to remember this information to validate the user. However, this responsibility is handled by Google Wallet, which keeps track of the value stored in the QR code once a user adds a card to their wallet app. Currently, the only information stored in the card is the student number, which is then compared with the database to verify whether the student is enrolled in the course and teaching session.

4.3 User Interface (UI)

The primary goal when designing the UI was to make a responsive UI that display the attendance data necessary for the user to know with as little effort as possible required by the user to get it. For instructors, the UI was designed to give a summary of each student's attendance history as a grade and whether the student is passing, and the instructor can click on a student's record to view the full history. The instructor can also see an even broader overview with two diagrams, one that shows the percentage of students meeting the attendance requirements, and one that shows that number of students that attended each week.

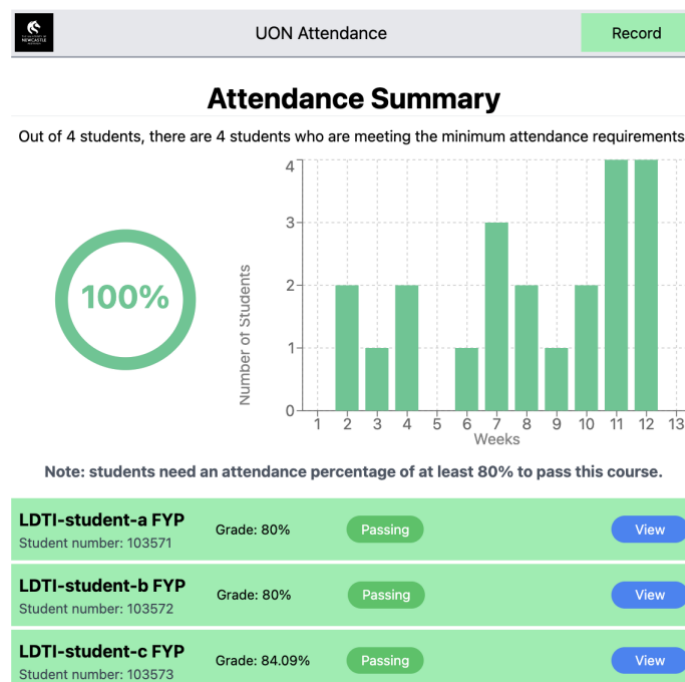


Figure 5: Instructor's view of attendance data

The view that instructors see when they click on a student's attendance record is almost identical to the view that a student sees when they open the application. The only main differences are that the instructor view specifies which student the attendance is for and gives an option to edit the attendance history. The full attendance history is displayed as a table on desktop devices, but once the screen is too small to display an entire table, it changes to a series of dropdowns for each week that initially show the percentage of classes attended, and once the user clicks on the dropdown, they see which specific classes were attended.

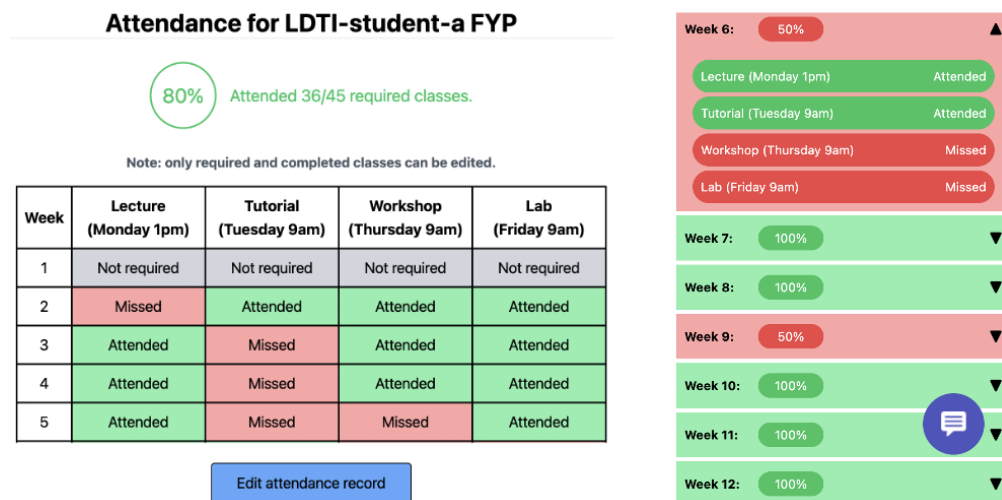


Figure 6: Full attendance history shown on desktop devices (left) and mobile devices (right)

Finally, the instructor can record attendance by using the record button in the navigation bar. The main way to record attendance is by scanning QR codes, which captures the student number stored as the value of the QR code and verifies the student with the backend. The application will stop scanning for two seconds and display a status message to the user, indicates whether the scan was successful or not. After two seconds, the application goes back to scanning cards.

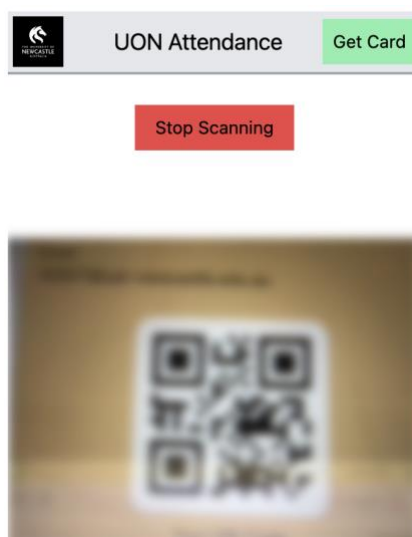


Figure 7: Instructor's view when scanning cards.

4.4 Authentication

Most of the authentication is done by Canvas when the user opens the application. Canvas first verifies whether the user is logged in as a valid user according to the records on Canvas, and then Canvas collects information on the user such as the type of the user, their name and ID, and the course that the LTI tool is being accessed from. Before sending this information to the application registered to the LTI tool, Canvas needs to verify the application by checking the private key of the LTI tool. Once this has happened, Canvas sends the user data to the application and displays the LTI tool to the user. When the application receives this data, it is stored so that any screen that would be displayed to a user checks whether the user has access to the screen, which typically just involves checking the type of user (student or instructor) and whether the screen that is being displayed is meant for that type of user.

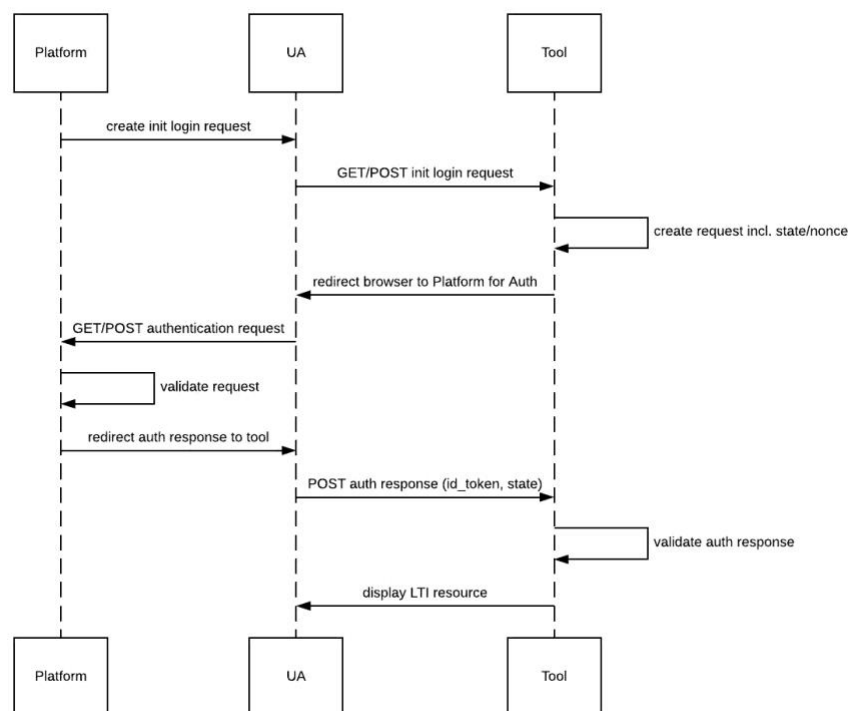


Figure 8: OpenID Connect Launch Flow (1EdTech Consortium, Inc., 2019)

3.8 Ethical Considerations

Canvas handles the user login process, which means that the application does not need to store sensitive user data such as passwords. When attendance records are updated, they are all logged with a timestamp to a log file, but the information included in the log file is only what is necessary, which includes the user ID and the attendance record ID. Since the application did not use real student data when testing, there wasn't as many ethical issues to consider like whether to encrypt student numbers or emails, but this would be something to consider before deployment. The University of Newcastle's code of conduct was adhered to for the entire development.

5. IMPLEMENTATION AND EVALUATION

This section outlines how the new attendance management system was implemented, including the development process, technologies used, and some challenges and solutions. Additionally, this section explains what the evaluation methods and criteria were, and what testing strategies were used.

5.1 Development Process

As explained in the methodology section, the approach used to develop this application was the agile methodology. It was completed in two main stages, the first being a proof-of-concept prototype that was developed to show that the core functions worked successfully, including:

- Allowing students to add a digital attendance card to Google Wallet.
- Allowing instructors to scan those cards.
- Updating the database whenever a student records their attendance.
- Integrating the attendance app with Canvas.

This prototype was created using HTML, CSS, and JavaScript on the frontend because of its simplicity, and Express.js on the backend with a PostgreSQL database. To integrate the application with Canvas, I used a local Docker image of Canvas for the development environment, which is mostly the same as the production environment but could only be accessed from one device. It allowed me to create test users and courses, which made initial development easy to start with lots of flexibility for testing.

The web application was stored in a GitHub repository so that no work was lost, and two branches were created for development. The first branch was the main branch and contained the application version from when it was last working. The second branch was the test branch, where I developed new features before merging it with the main branch. The prototype was completed and demonstrated to the client and stakeholders, before receiving approval to continue with the full attendance management system. Students were able to record their attendance and see their full attendance history on Canvas, however because the project had been developed using a monolithic architecture with no formal testing completed up to this point, it was decided that the application would be refined with a component-based architecture.

The component-based was made using React on the frontend, because it makes the code much easier to read and test. The list of attendance records could also be one component, which drastically reduced the amount of code needed as the code is mostly the same for the student and instructor views of attendance history. Tailwind CSS was also used rather than regular CSS, which meant each component contained its HTML, CSS, and JavaScript code in one file.

The development of the second prototype started by ensuring the application was integrated with a Canvas production server to more closely mimic the production environment. This required the authentication process to be designed once again, but this time security and access control was enforced securely by verifying the access tokens created during the LTI's launch. It also made it possible for other people to access the server with valid login details, so demonstrations were

much easier. However, since the PostgreSQL database was still local to one device, users on other devices could only view the application but not interact with it. Once it got closer to being completed, a mock database was set up that allowed the entire application to be used from any device with valid login details.

The database was redesigned and normalised, and some test data was also added. With the production server, I only had access to one course, two instructors, and four students for testing. After deciding to use the MVC design pattern, models were created to reflect tables in the database, which helped with maintaining data integrity by validating the data before recording it to the database, and controllers were created to split up the code and make it clear what the API endpoints were doing.

Each of the six database tables had two API endpoints for routing; the first was used to get all records in the table, and the second was to get a specific record using an ID. The only other API endpoint needed was for the attendance table, which needed a PATCH endpoint to update the status of a record when getting attendance. The application was then configured so that the routes on the server side could be called from the client side, and the attendance table was displayed on the home page of the application to ensure everything worked well. At this point, each route and model were unit tested using the Jest testing framework.

While the database and routing were being unit tested, the four core functions outlined in the first prototype were implemented again to work with React. Much of it was the same, except the code was much cleaner this time. The only change needed was to use a new library for scanning QR codes, as previously it had been generated in the `<script>` tag of the HTML document, but now it needed to be created with a `useEffect` React hook. After refining the implementation of the core features, the functions called were unit tested. A CI/CD pipeline was set up at the same time to run the previous unit tests again and catch any new bugs introduced.

The remaining additional features that had been left out from the first prototype were created next, which included:

- Allowing instructors to manually record student attendance.
- Allowing instructors to overwrite the attendance records if a mistake was made.
- Automatically updating the database each week by setting each attendance record's status for the current week to 'Missed' wherever the status was still NULL.
- Outputting user actions such as attendance recording and overwriting of attendance records to a log file with a timestamp.
- Creating a responsive UI that was displayed well on desktop and mobile.
- Creating diagrams for instructors to get an overview of course attendance and trends.

Once the additional features were implemented, they were unit tested and added to the CI/CD pipeline with the other unit tests. Finally, the entire system went through load testing to evaluate its performance before completing its development.

5.2 Challenges and Solutions

There were several challenges encountered during development of the new attendance management system. Firstly, integration with Canvas was difficult because the documentation for how to authenticate the LTI tool was not completely clear to understand. This was a hurdle that came up several times throughout development, but it never caused the project to be delayed because while the integration was being worked out, development on the core features continued.

Another challenge was when attempting to demonstrate and test the application with others. Not only was it impossible to have someone test the application on their own device for the first prototype, but even once a mock database was used with the production server it was still hard to test the software because users couldn't add a digital attendance card unless their email account was registered for testing. This a security feature by Google Wallet to make sure only approved accounts can add cards to their wallet app, but it meant that the solution to this problem was to send the user a screenshot of the card to be used for recording attendance. This worked for the project, but if we ended up using rotating barcodes, another solution would be needed.

The other problem that was unexpected with the new attendance management system was its performance when using lots of users at one time. This problem was not addressed before concluding development because it was discovered late when performance testing was being conducted, but it could be resolved in the future using horizontal scaling, caching, and more optimised strategies for retrieving data.

5.3 Evaluation Methods

Since the application was developed using the agile methodology, it only made sense to evaluate the prototype on a regular basis too. By evaluating the project frequently, we knew where the project was at and what to focus on in the future. Initially, the only evaluation method being used was through presentations to the project client and stakeholders, and the potential end-users of the tool. There were five presentations of the prototype made during the project's development:

- Prototype demonstration to client and stakeholders (23/05/2024). The aim of this presentation was to find out if the project was heading in the right direction. Specifically, whether the solution proposed met the project's requirements and would be considered a potential improvement over the current attendance tool. The prototype was demonstrated, and a Q/A session was completed at the end to receive feedback.
- Prototype demonstration to potential end-users (05/06/2024). Similar to the first demonstration but included both short and long-term goals in the future.
- Prototype presentation to client and stakeholders (29/07/2024). The purpose of this presentation was to give an update on the new prototype. It highlighted improvements made in maintainability, usability, and security. A Q/A session was completed at the end.
- Prototype demonstration to client and stakeholders (12/09/2024). The purpose of this demonstration was to show the changes made to improve the usability and readability of the application. A Q/A session was completed at the end.
- Prototype demonstration to potential end-users (21/10/2024). The final presentation was to report on the project's final status and what changes are needed for improvements.

Towards the final deadline of the project, usability and UAT tests were conducted to get real-time, active feedback from users as they spoke aloud while using the system. In each test, users were provided with four tasks to complete, and the time to complete them was recorded. At the end of each test, the user completed a questionnaire to give final written feedback. There were two test sessions conducted:

- Usability testing with two students (23/09/2024). The questionnaire used for this test was a custom questionnaire and the results are attached as appendix A and B.
- UAT with five potential end-users (28/10/2024). The questionnaire used for this test was the 10-question questionnaire used to calculate the System Usability Scale (SUS) score.

Finally, performance testing was completed using four repeated load tests on the application. The framework used for conducting the load tests was Artillery, which allows you to simulate many users at once and have them interact with the specified API endpoints.

5.4 Evaluation Criteria

The success criteria used in this project were clearly measurable and quantifiable so that it could be objectively stated whether it was met. Considerations such as time and cost are often considered to be the most important in a project, however for this project these were instead considered as project constraints. Instead, the success criteria defined in this project were used to measure functionality, performance, and user satisfaction, which are just as important (Agarwal & Rathod, 2006). These key deliverables were outlined in the project requirements and included:

- Quick attendance recording. The success criteria used to measure this was whether students could record their attendance in 15 seconds or less.
- Easy access to attendance history. The success criteria used to measure this was whether the attendance history could be viewed in 5 clicks or less. However, if you ignore the clicks potentially required to open and log in to Canvas LMS and select the course, then the success criteria becomes 2 clicks or less. It is better to ignore this, since the LTI tool developed is integrated with Canvas LMS, so these clicks are not a result of the project.
- Fast response time when viewing attendance history. The success criteria used to measure this was whether the attendance records loaded within 5 seconds of a student or instructor opening the application.
- Rapid updates of attendance records. The success criteria used to measure this was whether the attendance records were updated within 5 seconds of a student recording their attendance. This is important to ensure that the application is quick and ready to record another student's attendance after an appropriate duration.

It wasn't clear how many users would typically use the system at once, but given the fact that there are more than 10,000 students enrolled at the University of Newcastle, we estimated that having more than 100 users per second was possible to have, particularly at peak times when many students are attempting to record attendance. However, one of the advantages of using Google Wallet is that students don't need to interact with the application on their own device;

they only interact with it on an instructor's device. Despite this, we wanted the application to be able to handle at least 100 users per second without causing errors, and have a response time of less than 10 seconds.

5.5 Testing

Throughout the project, several testing approaches and methodologies were used rather than one specific method for the entire project. For the development of the first prototype, since it was a proof-of-concept prototype, the only testing performed was manual testing as the prototype was implemented. Once the development of the second prototype began, unit test cases were made on a regular basis, and the testing process was automated using a CI/CD pipeline to ensure the application was tested frequently as new features were created. This also helped to automate regression testing, as any new bugs that were introduced could be caught immediately without needing to create new test cases. The second prototype's approach to testing and evaluation meant that the agile methodology was kept consistent for development, testing, and evaluation.

5.4.1 Unit Testing

Unit testing, or module testing, is the lowest level of testing and tests small pieces of software that can be tested (Luo, 2001). For the project, it was conducted weekly once the main features of the second prototype had been developed, using the Jest testing framework. Units were tested on both the frontend and the backend, and included both black box testing and white box testing.

Black box testing is where software is tested based on the functional requirements without any knowledge of the internal structure or code (Sawant, Bari, & Chawan, 2012). Examples of black box tests used throughout the project included on input fields, such as when manually recording attendance. The input fields were tested to check whether the fields work correctly when using valid student details, and whether an error was returned when entering invalid details.

White box testing is the process of giving an input to the system and validating the output given (Sawant, Bari, & Chawan, 2012). It has the advantage of being able to use the source code to create test cases that might potentially cause unexpected errors. Examples of white box tests used throughout the project included using an invalid date or invalid week number for attendance.

Aside from testing the frontend with valid and invalid inputs, the backend units were also tested. The models were tested with both valid and invalid data inputs to ensure data integrity, prevent potential corruption of the database, and provide security against attacks such as SQL injections. The controllers were also tested for every API endpoint that used them for processing data.

Once the unit tests were created, a CI/CD pipeline was set up to run the tests whenever a push or merge commit was made on any branch. This helped automate the testing process, as a test suite could be automatically run against the code to ensure that the codebase remains stable without needing to retest everything (Orso & Rothermel, 2014). It offered a seamless way to automate regression testing, as regression testing often involves reusing the same tests again to test previously developed features after a new iteration. The application used GitHub Actions as the CI/CD tool, since the repository was also stored on GitHub, and it allowed us to make frequent

and continual updates to the software, and have reasonable certainty that the code did not introduce new bugs.

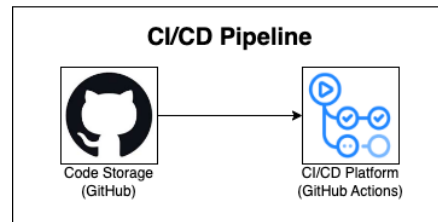


Figure 9: The application used GitHub Actions to run tests on GitHub.

The results from using the CI/CD pipeline were a success, as any time a commit was made that resulted in the unit tests failing, the commit was reversed, and the problem was fixed before trying again.

5.4.3 Usability and UAT Testing

Usability and UAT testing were not conducted formally until towards the end of development, because feedback from demonstrations was considered enough at earlier stages. Usability tests were first conducted a month earlier than the UAT tests because we wanted to get as much feedback and areas for improvement as possible from those and have enough time to implement them before final testing. The questionnaire used for the usability tests was much less formal and had more open-ended questions, with a particular emphasis on the UI, which features the user's like, which features they don't like, and which they wish were included. The only problem with the usability testing was that the sample size was very small, with only two users, and both users were students, so most of their feedback was coming from the student's perspective.

UAT testing used a more formal and structured approach by using the System Usability Scale, but we kept the same four tasks for the users to complete so that it was consistent, and the two user interfaces could be compared. Overall, the time to complete the tasks was similar, however the five users in the UAT testing on average gave more positive feedback on the UI. With an average SUS score of 94, the new attendance management system is considered very easy to use.

5.4.4 Performance Testing

Performance testing was completed so that the response times and success rate of the system could be measured when the system is being accessed by simultaneous users. The results indicate that at 100 users per second, the response time of the system is still quick with 0 errors or failures, which is acceptable for most cases where the system would be used. However, at 200 users, the response time becomes very long and 7% of users start to run into problems. The load tests only lasted a minute, so the number of errors is likely even higher as the 7% is calculated for the entire duration of the test, and errors only started occurring right at the end of the test.

6. RESULTS AND DISCUSSION

This section presents the results of testing the system and how effectively the new attendance management system meets the objectives of the project. There were three types of testing that were used to gain an insight into the final product's quality, which included usability testing, UAT testing, and load testing. The results from each test are displayed in a tabular format followed by an interpretation of the data that highlights what the system's strengths and weaknesses are, and which areas could be improved because of the tests.

6.1 Presentation of Data

The purpose of the usability tests was to get an early understanding and evaluation of the system's usability and overall satisfaction in terms of speed and user friendliness. The tests were completed by two students who had previously used the University of Newcastle's current attendance tool, and the key results are in Table 2. Each user was required to complete four tasks and answer a questionnaire at the end. The full questionnaire is also included in the appendix A which contains some open-ended questions too.

Table 2: Usability Testing Results (September 23)

Usability Criteria	Participant 1 Rating	Participant 2 Rating	Average
Completion Time	20 seconds	20 seconds	20 seconds
User Interface	3/5	3/5	3/5
Errors Encountered	0	0	0
Application Speed	5/5	5/5	5/5
Overall Satisfaction	3.5/5	4/5	3.75/5

The usability test results indicate that the application had a high performance with no errors encountered, but both users rated the user interface a 3/5 which suggested room for improvements in the design, particularly on mobile devices. This user interface rating was also what caused the completion time to be 20 seconds, which is generally acceptable, but the fourth task took on average 30 seconds to complete. Participant 1 recommended six changes and five of those were implemented before the next test, and participant 2 recommended three changes and all three were implemented before the next test.

Once the recommendations from the usability testing were implemented, a UAT phase was conducted to once again assess the usability of the system and the user's overall satisfaction. Five participants tested the final system using the same four scenarios as in the usability test. The System Usability Scale (SUS) was used to gain an insight into the overall satisfaction of the application and areas needing improvement. Verbal feedback throughout each test was also recorded for more specific criticism. The SUS scores are showed in Table 3 below.

Table 3: UAT Results (October 28)

Participant Number	SUS Score	Task Completion Rate	Task Completion Time
1	95/100	100%	15 seconds
2	100/100	100%	25 seconds
3	92.5/100	100%	30 seconds
4	100/100	100%	10 seconds
5	82.5/100	100%	20 seconds
Average	94/100	100%	20 seconds

The average SUS score across all five participants was 94, which indicates a high level of user satisfaction. All participants achieved a 100% task completion rate, although participant 3 took a while on some tasks. The average time to complete tasks was 20 seconds, which matches the time from the usability test. Overall, the results indicate that the new integrated attendance management system meets user expectations in terms of usability. It should be noted however that participant 5 was an instructor who has used the current attendance tool recently and would have preferred more control and options from the instructor's view.

After doing the usability test and getting a rating from each user on the application speed, it was realised that the application will always be fast with just one user, but with hundreds of users the application's performance will truly be put to the test. Hence, load testing was performed by changing the number of users getting attendance data every second. In total, four tests were completed, and the results are in Table 4.

Table 4: Load Testing Results

Test Number	Users per Second	Average Response Time	Throughput Requests	Error Rate
1	10	2.2 ms	30 requests/sec	0%
2	50	2.1 ms	150 requests/sec	0%
3	100	2.2 ms	300 requests/sec	0%
4	200	3386.7 ms	518 requests/sec	7%

The load testing results revealed that the system performs well under low to moderate loads (10 and 100 users per second) with no errors and an average response time of 2.2 ms. However, at high loads (200 users per second), the response time increases to over 3 seconds on average with 7% of the test users encountering errors. These results indicate that the system can handle fine under moderate loads but encounters scalability issues beyond 100 users per second.

6.2 Analysis and Interpretation

This section further analyses the results obtained from usability testing, UAT, and load testing, on the new attendance management system. The results show what objectives the new attendance system successfully achieves, and which areas need more improvement before deploying it for the entire university.

6.2.1 Usability Testing Results

The usability test results, which were performed on a previous version of the final system, indicate that attendance could be recorded in a very short time, with the average task taking on average 20 seconds. This time is also reduced when it comes to recording attendance for an entire class, since the page is already open and doesn't need to be closed and reopened again. The results also indicate that the new attendance management system resolves the issue with the current system of having invalid or incorrect attempts made when recording attendance.



Figure 10: Attendance Data Displayed on Desktop and Mobile Devices

However, because development of the new attendance management system was focused on the user interface on desktop devices, the display on mobile was a criticism from both users. The prototype was improved to have different user interfaces on each screen type, as seen in Figure 10. The summary of attendance data for instructors was also improved by adding two diagrams at the top of the screen, which allows instructors to easily see the trend in attendance over time.

6.2.2 UAT Results

The UAT results gave an average SUS score of 94 across five participants and all users were able to complete the task scenarios successfully. The average time to complete tasks still took 20 seconds like with the previous version, however this is because the tests were primarily done on desktop devices. It is also possible that some of the users in UAT had less familiarity with the current attendance system and technology in general, as both users in the usability test were students who had seen a previous prototype before the usability test.

While there were no errors encountered during the tests, there were some improvements recommended by multiple users that should be made to the application before deployment, including:

- Add 'back' buttons when navigating away from the home page.

- Merge manual attendance recording and overwriting attendance together since they are both similar in use case.
- Make the buttons on the navigation bar more obvious that they can be clicked.
- Give instructors more control over getting attendance data and changing the data.
- Add a third user type for teaching assistants and lab demonstrators, who have access to less information but can still record attendance for students.
- Allow students to record attendance for a class even if they are not enrolled in it.

Overall, the results show that the new attendance method of using QR codes is very user-friendly and easy to use, but there are some missing features that were not included in the scope of this project but would need to be added before using the attendance management system in practice.

6.2.3 Load Testing Results

The load testing results reveal critical problems regarding the scalability of the system. It seems that below 100 users, the system can handle okay without any significant change in performance, but above a certain threshold (e.g. 200 users) the load results in multiple errors and skyrocketing response times. Overall, the indicates that the system currently would handle fine during normal activity, but at peak periods with many students recording attendance, such as at the start of classes, the attendance management system will struggle. This means that the system needs some method to increase the scalability of the system, which could be from caching attendance data, reducing the number of API calls if possible, or using horizontal scaling after deployment.

6.3 Contributions

QR code-based attendance systems are commonly used when it comes to recording attendance for academic institutions. However, most of the previous solutions that implement QR code-based attendance systems coupled them with user accounts, biometrics and/or GPS tracking to solve the issue of preventing fraud. The attendance system that was implemented in this project aimed to fill this gap in the literature by still using QR codes without any additional factors, but use QR codes that switch the encoded value at a set interval. Typically this would need to be done by switching the value stored in a database and checking whether it matches. Our attendance system uses Google Wallet to create digital attendance cards with QR codes on them, which has a feature to rotate the barcodes periodically. However, this did not end up getting implemented in time, so more work needs to be done before giving a conclusion as to whether this feature solves the problem of fraud at a satisfactory level.

6.4 Comparison with Literature

QR code-based attendance systems in recent literature have often been coupled with features such as user authentication, biometric verification, or GPS tracking. For example, (Liew & Tan, 2021) tracked the student's locations once they scanned the QR code, (Masalha & Hirzallah, 2014) went even further by using user authentication, biometric verification, and GPS tracking to

enhance fraud prevention, and (Wenny et al., 2022) used no additional authentication factors but recommended using something like facial recognition to address the problem.

In comparison, the new attendance system covered in this report aims to prevent fraud without making the implementation much more complicated by using a hybrid attendance method. The use of the in-built feature offered by Google Wallet to auto-generate rotating QR codes aims to address the problem in an easy and simple way. However, since the approach was never tested to check how well it prevents fraud compared to other attendance methods, further work is required before a definitive conclusion can be made on the effectiveness of our chosen solution.

6.5 Limitations

There are several limitations with the proposed solution:

- **Dependence on Google Wallet.** Since the digital attendance cards are generated by Google Wallet, the new attendance system can only be used by students with Android devices. This can be addressed by either allow Apple Wallet cards to be scannable too, or by having the QR codes generated by the university itself, however this approach requires the QR code information to be stored by the university which requires a lot of storage.
- **Implementation of rotating QR codes.** Due to time constraints, this feature didn't end up getting implemented, so it is hard to draw a conclusion about the system's effectiveness in preventing fraud.
- **Missing features.** The proposed solution does not have all the administrative features that were present in the current attendance management system, which instructors will want to be included in the new attendance management system.
- **Scalability and load testing.** The proposed solution was not tested on a university scale, but the load tests showed that the system cannot handle huge amounts of users simultaneously. Any future work should focus on optimising the system's performance under high traffic.
- **Small sample size.** The number of users that completed usability and UAT testing was very small, so a larger sample size would give a more confident conclusion that the system is easy to use and does not have any errors when recording attendance.

7. CONCLUSION

While there were several good options for the implementation of the attendance management system, we went for the QR code-based attendance management system with an innovative alternative to preventing fraud that most other attendance management systems have not went for. The main purpose was to create a new attendance management system for the University of Newcastle that would have great usability and resolve many of the problems initially recognised with their current attendance management system. The following section summarises the key findings, implications, and recommendations from this research.

7.1 Summary

This project successfully demonstrated the feasibility of using an integrated attendance management system that records attendance by scanning QR codes, without needing to complicate the implementation and affect usability with an extra authentication method like location tracking which was the problem with the current attendance tool. The results from usability and UAT tests indicate that users can complete the tasks in a reasonable amount of time, and they find the new system easy to use. However, the system currently suffers from scalability issues when dealing with more than 100 users per second, and time constraints led to the rotating QR codes feature offered by Google Wallet not being implemented in time. There are also some missing features that would be good to include, particularly for instructors, so more implementation and optimisation is required before deployment to be used by all students.

7.2 Implications

This project contributes to existing literature by attempting to use modern technology like rotating QR codes to address the problem of preventing fraud when just using QR codes by themselves. The success of this technique will allow for low-maintenance and low-cost attendance management systems to be easily developed in the future for academic institutions, without needing complex or inconvenient methods like biometrics. The project has also shown that QR code-based attendance management systems are practically free from errors, unlike the use of GPS location tracking that is currently used by the University of Newcastle.

7.3 Recommendations

Future research should focus on implementing the rotating QR code feature either with or without Google Wallet to see how much it deters fraud and makes it difficult to record attendance without being in the classroom. Once this has been completed, the system should be improved so that it can be integrated with not just Android devices, but anyone with a phone. Alternatively, the system could be improved so that physical cards can also still be used in case a student forgot their phone. Finally, the system needs to be more scalable so that it can be used in the real-world with thousands of users.

REFERENCES

- 1EdTech Consortium, Inc. (2019, May 15). *1EdTech security framework v1.0*. IMS Global Learning Consortium. <https://www.imsglobal.org/spec/security/v1p0/>
- Adewole, K. S., Abdulsalam, S. O., Babatunde, R. S., Shittu, T. M., & Oloyede, M. O. (2014). Development of fingerprint biometric attendance system for non-academic staff in a tertiary institution. *Development*, 5(2), 62-70.
- Ali, N. S., Alhilali, A. H., Rjeib, H. D., Alsharqi, H., & Al-Sadawi, B. (2022). Automated attendance management systems: systematic literature review. *International Journal of Technology Enhanced Learning*, 14(1), 37-65.
- Arif, Z. H., Ali, N. S., Zakaria, N. A., & Al-Mhiqani, M. N. (2018). Attendance management system for educational sector: critical review. *International Journal of Computer Science and Mobile Computing*, 7(8), 60-66.
- Dixon, J., & Abuzneid, A. S. (2020, December). An NFC based student attendance tracking/monitoring system using an IoT approach. In *2020 International Conference on Computational Science and Computational Intelligence (CSCI)* (pp. 1082-1086). IEEE.
- Eng, M. S. (2022). *Digital student ID card using NFC technology* (Doctoral dissertation, UTAR).
- Kadry, S., & Smali, M. (2010). Wireless attendance management system based on iris recognition. *Scientific Research and Essays*, 5(12), 1428-1435.
- Leal, J. P., Queirós, R., Ferreira, P., & Swacha, J. (2022). A roadmap to convert educational web applications into LTI tools. In *Third International Computer Programming Education Conference (ICPEC 2022)*. Schloss-Dagstuhl-Leibniz Zentrum für Informatik.
- Lewis, J. R. (2018). The system usability scale: past, present, and future. *International Journal of Human-Computer Interaction*, 34(7), 577-590.
- Liew, K. J., & Tan, T. H. (2021, September). QR code-based student attendance system. In *2021 2nd Asia Conference on Computers and Communications (ACCC)* (pp. 10-14). IEEE.
- Lim, T. S., Sim, S. C., & Mansor, M. M. (2009, October). RFID based attendance system. In *2009 IEEE Symposium on Industrial Electronics & Applications* (Vol. 2, pp. 778-782). IEEE.
- Masalha, F., & Hirzallah, N. (2014). A students attendance system using QR code. *International Journal of Advanced Computer Science and Applications*, 5(3).
- Moshayedi, A. J., Roy, A. S., Liao, L., Lan, H., Gheisari, M., Abbasi, A., & Bamakan, S. M. (2021). Automation attendance systems approaches: a practical review. *BOHR Int. J. Internet Things Artif. Intell. Mach. Learn*, 1, 23-31.
- Murray, J., & Feinberg, D. (2020). Collaboration and integration: embedding library resources in Canvas. *Information Technology and Libraries*, 39(2).
- Qureshi, M. (2020). The proposed implementation of RFID based attendance system. *International Journal of Software Engineering & Applications (IJSEA)*, 11(3).
- Rokade, Y., Shetty, P., & Wankhade, D. S. (2017). Automated attendance system using NFC. *IJARIIIE*, 3(3), 1-6.

Sawant, A. A., Bari, P. H., & Chawan, P. M. (2012). Software testing techniques and strategies. *International Journal of Engineering Research and Applications (IJERA)*, 2(3), 980-986.

Shin, D. H. (2009). Towards an understanding of the consumer acceptance of mobile wallet. *Computers in Human Behavior*, 25(6), 1343-1354.

Singh, M., Khan, M. A., Singh, V., Patil, A., & Wadar, S. (2015, February). Attendance management system. In *2015 2nd International Conference on Electronics and Communication Systems (ICECS)* (pp. 418-422). IEEE.

Talip, B. A., & Zulkifli, M. Z. (2018). Mobile attendance system using QR codes technology. *Journal of Computing Technologies and Creative Content*, 3(1), 1-3.

APPENDIX A. USABILITY TEST QUESTIONNAIRE 1

Tasks

1. As a student, find out whether you are meeting the attendance requirements. If not, find out what your attendance grade is.
2. As a student, record your attendance for a class (it is scheduled to occur now). You will be given a laptop with an instructor signed in to use to record attendance. Check whether your attendance was recorded for the current class, and whether the overall attendance grade has been updated.
3. Assume you have forgotten your phone and cannot record attendance as before. As an instructor, record the attendance manually, and check whether the status has been updated from the student's perspective.
4. Suppose a student attended class but forgot to record their attendance. As an instructor, overwrite the attendance record status and check whether it has been updated from the student's perspective.

1. Task Completion

Task 1) Successfully completed in 10 seconds. Difficulty: **1/5**

Task 2) Successfully completed in 20 seconds. Difficulty: **1/5**

Task 3) Successfully completed in 30 seconds. Difficulty: **1/5**

Task 4) Successfully completed in 20 seconds. Difficulty: **3/5**

2. User Interface

- Rate the user interface on a scale of 1-5.
- Was there anything confusing about it?
- Rank the following user interface designs for both mobile and desktop views.

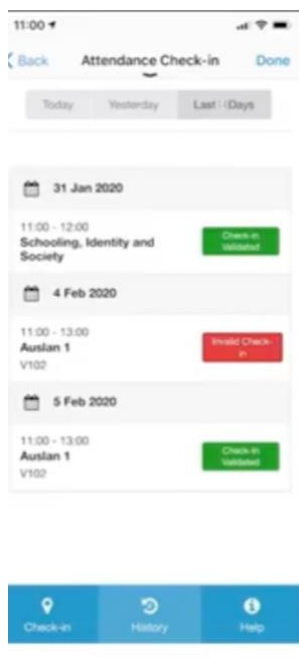


Figure 1: Current

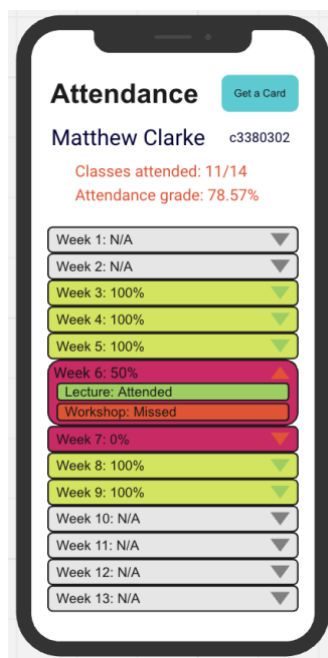


Figure 2: Alternative

Summary for LDTI-student-a FYP

Attendance percentage: 74.19%
 Attended 23/31 required classes.

#	Lec	Tut	Lec	Wor	Lab
1	N/A	N/A	N/A	N/A	N/A
2	Y	N	Y	Y	Y
3	Y	Y	Y	Y	Y
4	Y	Y	Y	Y	Y
5	Y	Y	Y	Y	Y
6	N	Y	Y	N	N
7				N	
8	N	Y	Y	N	N
9					
10					
11					
12					
13					

Figure 3: New

74.19% Attended 23/31 required classes.

Week	Lecture (Mon)	Tutorial (Tue)	Lecture (Wed)	Workshop (Thu)	Lab (Fri)
1	Not required	Not required	Not required	Not required	Not required
2	Attended	Missed	Attended	Attended	Attended
3	Attended	Attended	Attended	Attended	Attended
4	Attended	Attended	Attended	Attended	Attended
5	Attended	Attended	Attended	Attended	Attended
6	Missed	Attended	Attended	Missed	Missed
7				Missed	
8	Missed	Attended	Attended	Missed	Missed

Figure 4: New

Attendance				Get a Card
Matthew Clarke	c3380302	Classes attended: 10/14	Attendance grade: 71.43%	
Week 1: 100%	Week 2: 100%	Week 3: 100%		
No classes	No classes	Lecture: Attended Workshop: Attended		
Week 4: 50%	Week 5: 100%	Week 6: 100%		
Lecture: Attended Workshop: Missed	Lecture: Attended Workshop: Attended	Lecture: Attended Workshop: Attended		
Week 7: 0%	Week 8: 100%	Week 9: 50%		
Lecture: Missed Workshop: Missed	Lecture: Attended Workshop: Attended	Lecture: Attended Workshop: Missed		

Figure 5: Alternative

Feedback:

- 4/5.
- Have a required attendance there so the student doesn't need to go and find it.
- Table and use of colour is nice. Maybe use ticks and crosses.
- Have a better way of letting the user know the card was scanned – maybe don't show the camera on again.

- Table rows weren't immediate obvious to click.
- Edit button wasn't viewable immediately.
- For phones: figure 3 is preferred because all the data is viewable immediately.
- For desktops: definitely figure 4 is preferred over figure 5.

3. Errors

- Were there errors?

Feedback:

- No errors.

4. Application Speed

- Rate the application speed on a scale of 1-5.

Feedback:

- 5/5.

5. Features

- Which features did you find particularly useful?
- Which features were not useful?
- Which features would you like to have seen included?

Feedback:

- The table was cool. Scanning was good. It is quick to scan cards too.
- Everything had a purpose.
- No other features needed.

6. Final Evaluation

- What is your overall evaluation for this application on a scale of 1-5?
- Would you prefer this attendance tool over the current one?
- Is there any other feedback you would like to give?

Feedback:

- 3.5/5.
- I do prefer this attendance tool over the current one.
- The phone UI could be improved.

APPENDIX B. USABILITY TEST QUESTIONNAIRE 2

Tasks

1. As a student, find out whether you are meeting the attendance requirements. If not, find out what your attendance grade is.
2. As a student, record your attendance for a class (it is scheduled to occur now). You will be given a laptop with an instructor signed in to use to record attendance. Check whether your attendance was recorded for the current class, and whether the overall attendance grade has been updated.
3. Assume you have forgotten your phone and cannot record attendance as before. As an instructor, record the attendance manually, and check whether the status has been updated from the student's perspective.
4. Suppose a student attended class but forgot to record their attendance. As an instructor, overwrite the attendance record status and check whether it has been updated from the student's perspective.

1. Task Completion

Task 1) Successfully completed in 10 seconds. Difficulty: **3/5**

Task 2) Successfully completed in 10 seconds. Difficulty: **1/5**

Task 3) Successfully completed in 20 seconds. Difficulty: **1/5**

Task 4) Successfully completed in 40 seconds. Difficulty: **4/5**

2. User Interface

- Rate the user interface on a scale of 1-5.
- Was there anything confusing about it?
- Rank the following user interface designs for both mobile and desktop views.

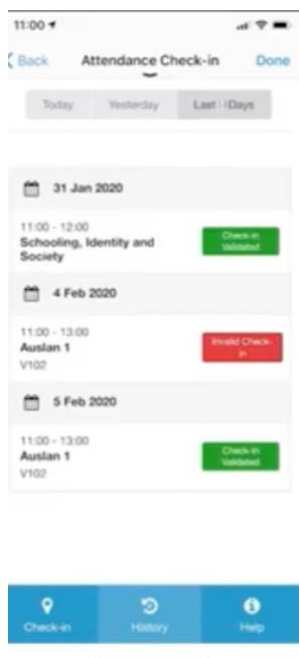


Figure 1: Current

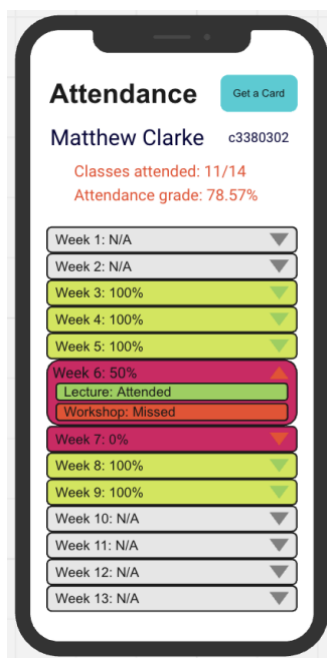


Figure 2: Alternative

Summary for LDTI-student-a FYP

Attendance percentage: 74.19%
 Attended 23/31 required classes.

#	Lec	Tut	Lec	Wor	Lab
1	N/A	N/A	N/A	N/A	N/A
2	Y	N	Y	Y	Y
3	Y	Y	Y	Y	Y
4	Y	Y	Y	Y	Y
5	Y	Y	Y	Y	Y
6	N	Y	Y	N	N
7				N	
8	N	Y	Y	N	N
9					
10					
11					
12					
13					

Figure 3: New

74.19% Attended 23/31 required classes.

Week	Lecture (Mon)	Tutorial (Tue)	Lecture (Wed)	Workshop (Thu)	Lab (Fri)
1	Not required	Not required	Not required	Not required	Not required
2	Attended	Missed	Attended	Attended	Attended
3	Attended	Attended	Attended	Attended	Attended
4	Attended	Attended	Attended	Attended	Attended
5	Attended	Attended	Attended	Attended	Attended
6	Missed	Attended	Attended	Missed	Missed
7				Missed	
8	Missed	Attended	Attended	Missed	Missed

Figure 4: New

Attendance				Get a Card
Matthew Clarke	c3380302	Classes attended: 10/14	Attendance grade: 71.43%	
Week 1: 100%		Week 2: 100%		Week 3: 100%
No classes		No classes		Lecture: Attended Workshop: Attended
Week 4: 50%		Week 5: 100%		Week 6: 100%
Lecture: Attended Workshop: Missed		Lecture: Attended Workshop: Attended		Lecture: Attended Workshop: Attended
Week 7: 0%		Week 8: 100%		Week 9: 50%
Lecture: Missed Workshop: Missed		Lecture: Attended Workshop: Attended		Lecture: Attended Workshop: Missed

Figure 5: Alternative

Feedback:

- 3/5 – could use a component library such as material ui or chakra ui to make things look much smoother – overall layout was standard.
- The table on the instructor's side was not obviously clickable.
- The navbar was good, it was clear what it did

- I prefer figure 2 for the mobile screens mostly because it shows the summarised version of the data. Figure 1 is my least favourite one.
- I prefer figure 4 for the desktop screens as it shows all the data – anyone accessing on a desktop is going to want to see the complete dataset.

3. Errors

- Were there errors?

Feedback:

- No errors.

4. Application Speed

- Rate the application speed on a scale of 1-5.

Feedback:

- 5/5.

5. Features

- Which features did you find particularly useful?
- Which features were not useful?
- Which features would you like to have seen included?

Feedback:

- The full table and being able to edit as the instructor were very good.
- No features were useless – everything had a purpose.
- I would have liked to see a collapsible table.

6. Final Evaluation

- What is your overall evaluation for this application on a scale of 1-5?
- Would you prefer this attendance tool over the current one?
- Is there any other feedback you would like to give?

Feedback:

- 4/5.
- I do prefer this attendance tool purely because of the display of all the data.
- No other feedback.